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Setting Up Gaze Mapping

Gaze Map Analysis maps eye tracking data onto a 'static' reference images that appears withing a 'dynamic' stimulus recording, such as screen/website recordings or eye tracking glasses camera recordings. This article explains the Gaze Map interface and walks through the steps for selecting reference images and using Gaze Map Analysis. These steps are also described with some additional information in this article about [Gaze Mapping](#).

When to use Gaze Map Analysis?

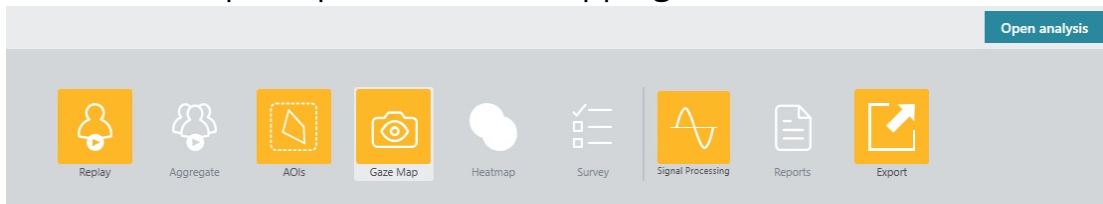
- Gaze Map Analysis can be performed on any **dynamic stimulus** that includes eye tracking data. This includes videos media or recordings - e.g. screen/website recording, eye tracking glasses, or camera recordings.
- Gaze Map Analysis will map eye tracking data onto a **static reference image** that appears in the dynamic recording. Reference images are selected in the Gaze Map window, where analysis is also performed.
- Reference Images can be created from the stimulus or any image that shows where gaze mapping will be applied.
- The Gaze Map scenes (resulting from Gaze Map Analysis) allow Aggregate analyses that are only available for static stimuli, such as **Heatmaps** and **Scan Paths**.
- Gaze Map Analysis does not apply (and is not available) for static stimuli - e.g. images or surveys.

Gaze Mapping Editor

The Gaze Mapping Editor can be found in any analysis that includes dynamic stimuli and eye tracking data.

Opening the Gaze Mapping Editor

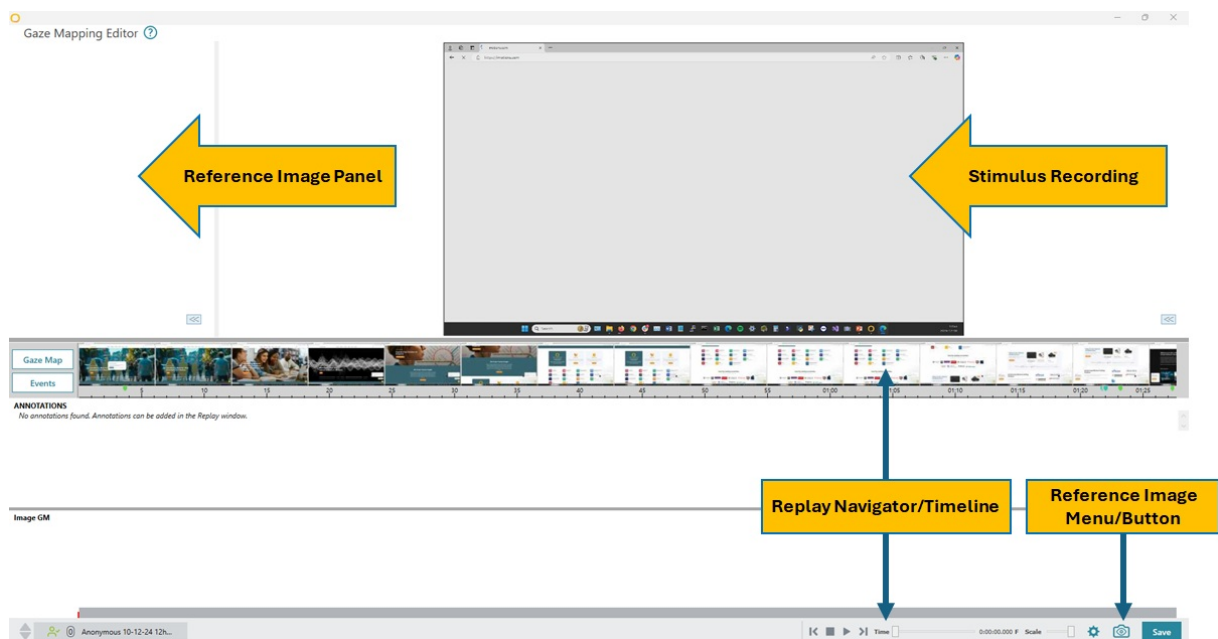
1. Create or select an analysis that includes (at least one) dynamic stimulus.
2. Select a dynamic stimulus (Gaze Map is unavailable/grayed out for static stimuli).
3. Click Gaze Map to open the Gaze Mapping Editor.



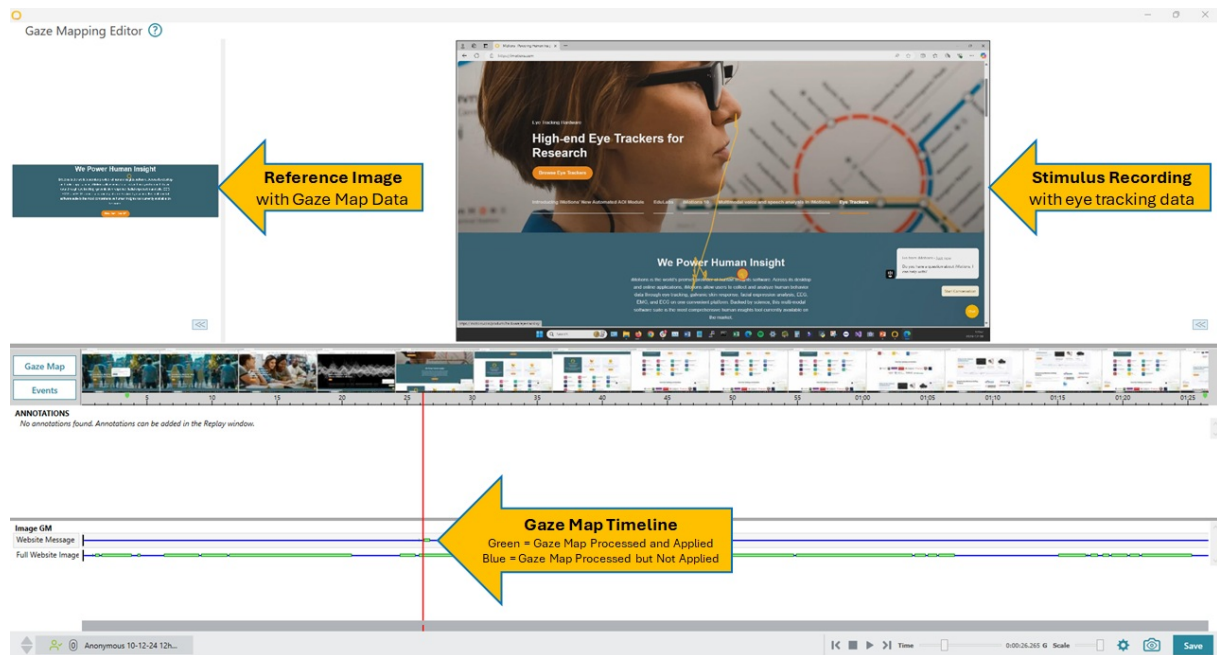
Gaze Mapping Editor Window

The Gaze Mapping Editor is used to create, view, and edit Gaze Maps.

- **Reference Image Panel** displays the selected Gaze Map.
- **Stimulus Recording** shows the stimulus replay.
- You can navigate through the replay with the playback buttons or the timelines in the middle or bottom of the window.
- **Reference Image Menu** is used to select reference images and create Gaze Maps.



Gaze Mapping Editor - New Analysis before Gaze Mapping



Gaze Mapping Editor with Processed Gaze Maps

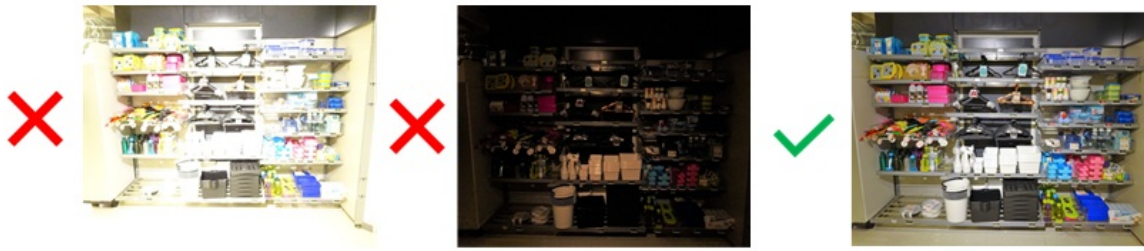
Selecting Reference Images

Reference Images can be selected from image files or video frames from any dynamic stimulus - Gaze Mapping will overlay eye tracking data onto Reference Images.

Recommendations for reference images

Reference Images should clearly show relevant visual objects/features for your analysis. Reference image quality is critical for Gaze Mapping, especially for real-world recordings/images.

- Reference Images should clearly show the visual objects/features where Gaze Mapping will apply.
- Resolution and lighting should be sufficient to provide a sharp image, avoiding overexposed (too bright) or underexposed (too dark) images.
- Objects/features should appear as expected in the dynamic stimulus (e.g. store shelves that are centered and aligned with recorded view).



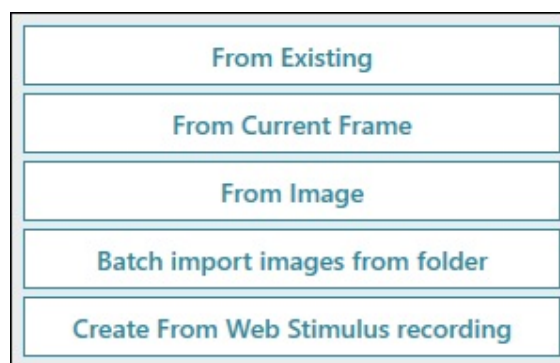
Example Reference Images (optimal lighting in last image)

Creating Gaze Maps

Click on the Reference Image menu/button (at the bottom-right of the Gaze Mapping Editor).

Select Reference Image(s) with one of the menu options (hover with mouse cursor over any option for additional information).

- **From Existing** provides a list of existing Gaze Maps, which can be applied to the currently selected recording.
 - Selecting 'From Existing' Gaze Maps will aggregate data across respondents (e.g. for AOI analyses, Heatmaps, and Replays of the Gaze Mapped data on the reference image).
 - We recommend reviewing the resulting Gaze Map before selecting and processing it across respondents (described below).
- **From Current Frame** will apply the frame shown in the Gaze Mapping Editor as a reference image.
- **From Image** allows selection of an image file.

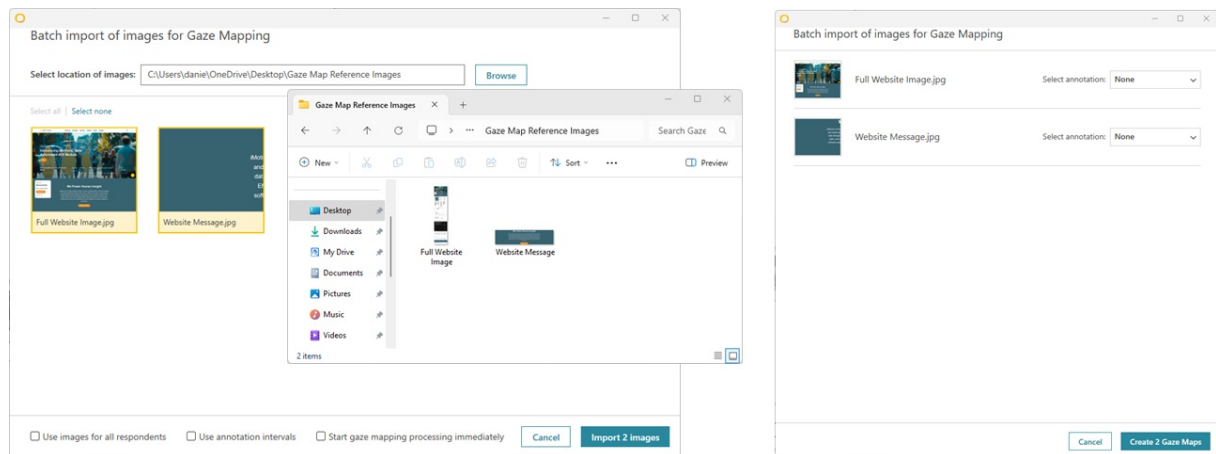


Reference Image Menu

Batch image import from folder

Multiple Reference Images can be imported and selected for Gaze Mapping (from any folder containing image files).

This also allows you to use Annotations to define Gaze Map intervals for each reference image.

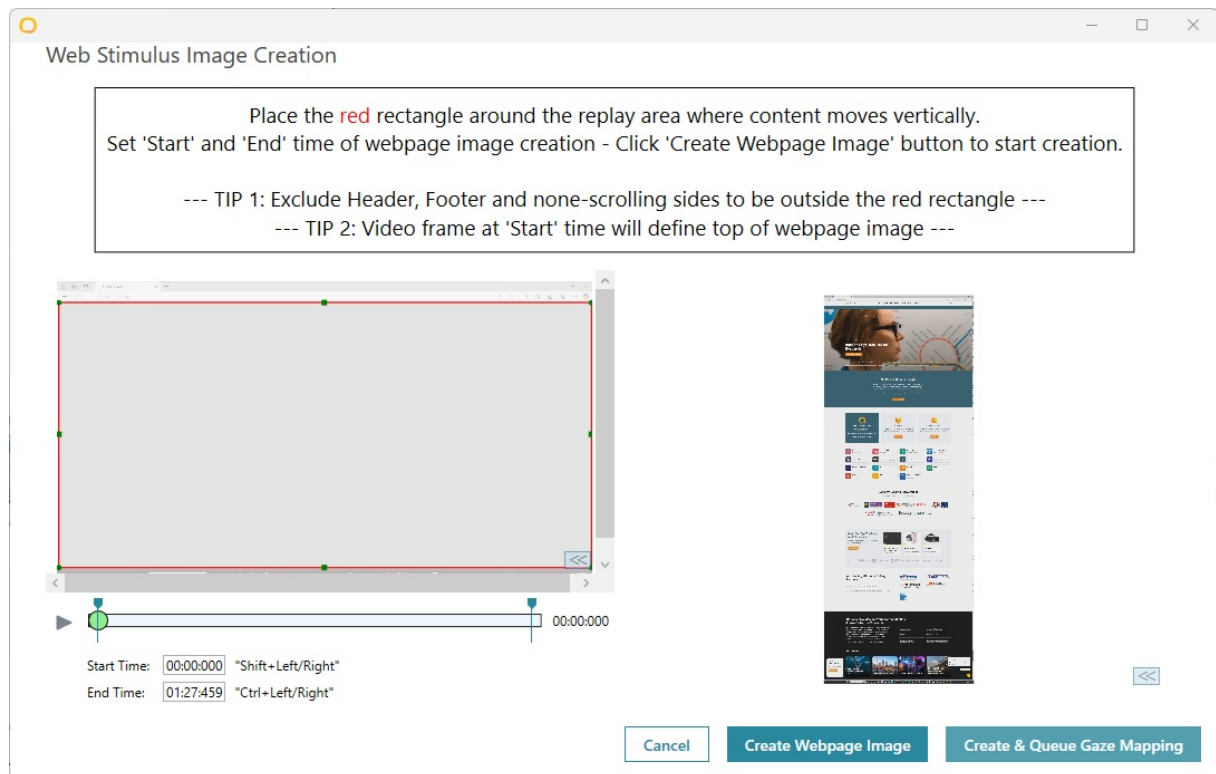


Batch Import Reference Images

Create from Web Stimulus recording

A Reference Image can be created from any web stimulus recording - providing a 'full page' reference image of the website.

Instructions are provided in the Web Stimulus Image Creation interface.



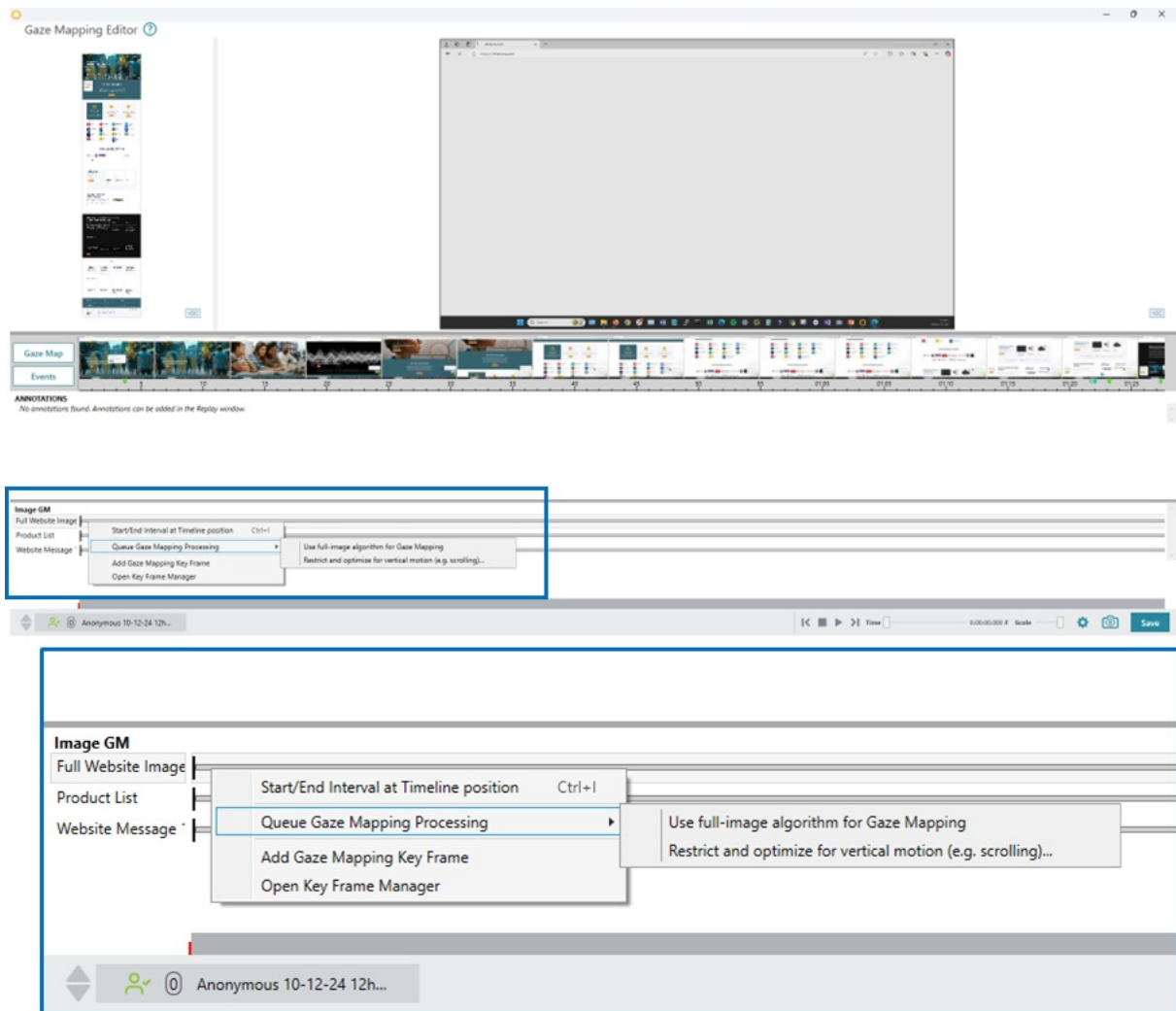
Create Webpage Image from Stimulus Recording

Adjusting Gaze Map Intervals

Gaze Map Intervals in the **Image GM timeline** to control where processing is applied (and reduce the time needed for processing).

Intervals can be set at the selected time point by:

- Pressing **Control+I** key
- Right-clicking on the interval and then clicking **Start/End Interval at Timeline Position**.
- Clicking and dragging interval start/end points
- Annotations can also be used to define gaze map intervals.



Gaze Map options (right-click on Image GM timeline) to adjust intervals and queue processing

Running Gaze Map Processing

Gaze Map processing can be optimized by adjusting Gaze Map Intervals, selecting the appropriate Gaze Map algorithm, and reviewing Gaze Map outputs, and (if necessary) Keyframe Corrections.

Gaze Map processing can be initiated from several locations:

- Image GM timeline (right-clicking on the interval and queuing processing)
- Options in Reference Image selection steps for 'Batch import' and 'Web Stimulus recording'

- The main iMotions window (right-clicking on the study and selecting Post Processing>Gaze Mapping).

There are two algorithms for Gaze Map Processing:

- Standard **'Full-image' Algorithm**
 - Applies to eye tracking glasses and most stimulus recordings
- Optional Algorithm that will **'Restrict and optimize for Vertical Motion (e.g. Scrolling)'**
 - Applies to some website stimulus recordings where visual content move with the browser window (e.g. website banners, borders)

Use full-image algorithm for gaze mapping

The *standard algorithm* for Gaze mapping is this *Full-image algorithm for gaze mapping*. This algorithm involves an object recognition algorithm that searches for the visual features of the 'reference' image in the original recording.

If the algorithm is able to localize the 'reference' image in the eye tracking recording, this algorithm maps the original gaze-points onto their corresponding location in the 'reference' image.

Some examples when Gaze mapping would not be applied: (1) the 'reference' image is not present in the video recording made by mobile eye tracking glasses because the respondent looked away from the frame, (2) the 'reference' image is present in the video recording, but the respondent does not look inside the frame of the 'reference' image (i.e., the gaze-point falls outside of the 'reference' image), (3) the 'reference' image is not similar enough to the original recording environment, or (4) the quality of the 'reference' image and video recording are not similar enough (e.g. blurred video-frames due to fast movements).

Restrict and optimize for vertical motion (e.g. Scrolling)

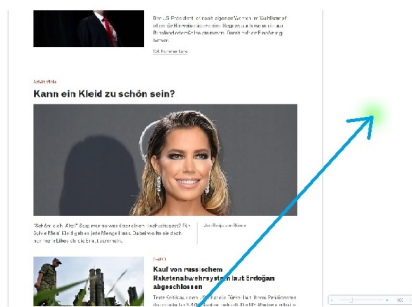
The *alternative algorithm* for Gaze mapping is the *Restrict and optimize for vertical motion (e.g., scrolling, ...) algorithm*. This algorithm is mostly relevant for websites; in particular, websites with side banners that do not move together with the rest of the website when scrolling down ('vertical motion').

In such a case, the Restrict and Optimize for vertical motion (e.g., scrolling, ...) option allows you to define what part of the website is dynamic (i.e., the information changes when scrolling down) and what part of the website is static (i.e., the information does not change when scrolling down). This is done by placing a square shape over the *dynamic* part of the website; this is typically the central part of the website (excluding menus at the top, side banners, cookies at the bottom of the screen, etc.). The algorithm then automatically applies a Gaze mapping routine in two waves; a first Gaze mapping routine for the dynamic part of the website (assuming vertical motion), and a second Gaze mapping routine for the static part of the website (assuming the absence of vertical motion).

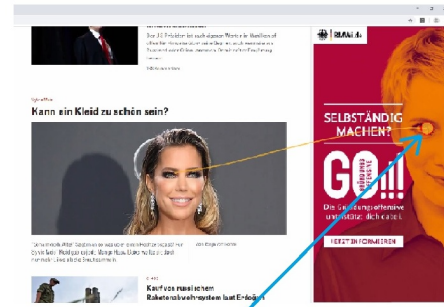
When to use the algorithm optimized for vertical motion?

This alternative algorithm should be used when a website or screen recording, contains visual content that remains in a location within the browser window when scrolling occurs (e.g. ad banner on the side, menu bar at the top).

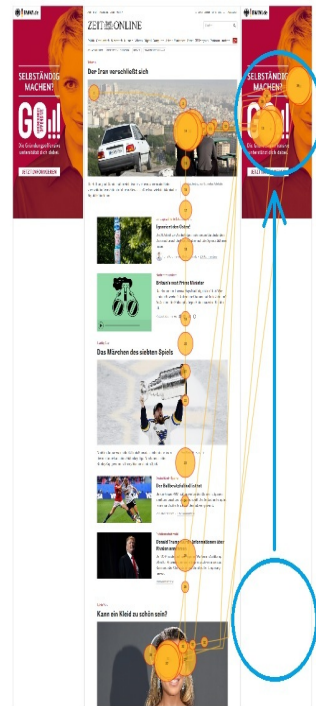
The following Gaze Map screenshots show an example when the standard algorithm maps gaze data from the recording onto an inaccurate location in the reference image. In this case, the gaze point should be mapped on the ad banner. This occurred because the ad banner would remain along the right of the browser window while scrolling down the website. Using the alternative algorithm will correctly overlay gaze points on the ad banner, regardless of the vertical location on the website.



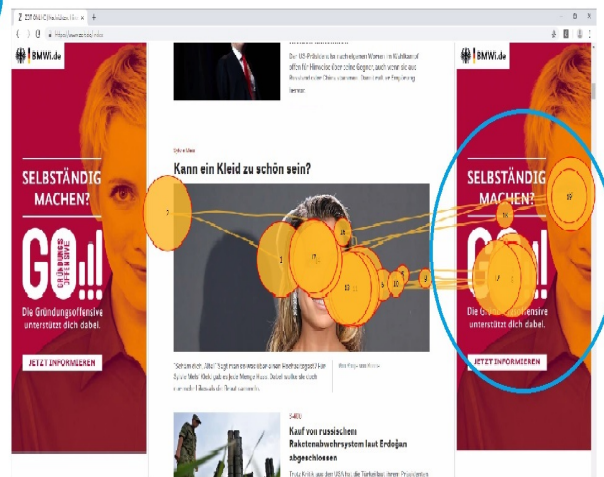
Gaze point gets mapped in relation to the picture that was next to it, but does not get remapped onto the ad. So it is difficult to analyse where on the ad the respondent was focusing on if the ad changed its position relative to the other content.



In the replay of the recording, one can see that the respondent looked at the ad.



Remapped gaze on reference image

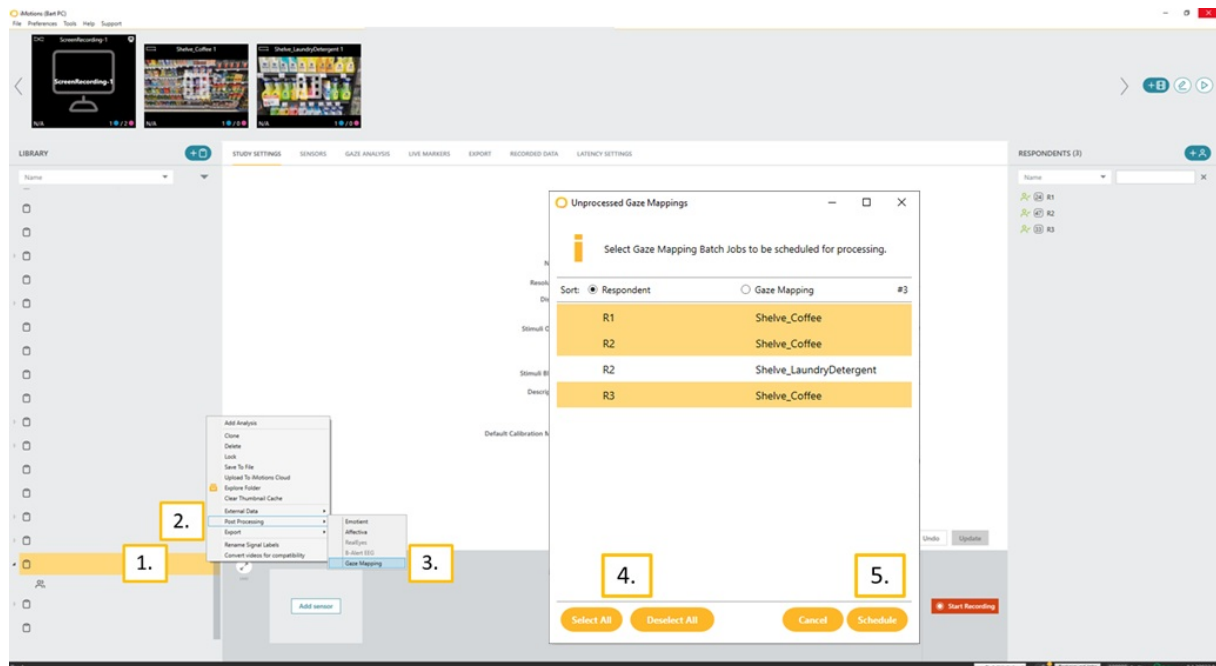


Original gaze replay in recording

Batch Post Processing

Batch Post Processing allows you to process a group of respondents and gaze maps that have been set up, but not yet processed.

1. Select a study that includes unprocessed gaze maps.
2. Right-click on the study and select **Post Processing**
3. Select **Gaze Mapping**.
4. Select from list of **Unprocessed Gaze Mappings**.
5. Click **Schedule** to initiate processing.



Batch Gaze Map Processing

Cloud Batch Post Processing

Gaze Map processing can be performed with iMotions Cloud Batch Processing, rather than local processing on your computer.

Cloud Batch Processing is first uploaded to the iMotions Cloud, where it is processed and sent back to your computer. Gaze mappings files are never stored in the cloud and returned immediately to your computer. To facilitate processing of multiple gaze maps, uploaded video materials are temporarily retained in the cloud (for maximally 30 days).

Important Note: This requires the **license module for Cloud Batch Processing**.

How to Enable and Use Cloud Batch Processing

1. Open iMotions **Global Settings** ('Preferences' menu or Alt+G key)
2. Select the 'Results' tab and tick the checkbox for **'Remote gaze mapping batch processing (restart required)'**
3. Review information about data usage and click 'Ok'.
4. Click 'OK' to save Global Settings.
5. Restart iMotions and initiate Batch Processing (with the steps described in the preceding section).

Reviewing Gaze Maps and Applying Key Frames

Gaze Maps can be reviewed in the Gaze Mapping Editor. When necessary, Gaze Maps can be refined with Key Frames - selecting stimulus frames and connection points to reference images.

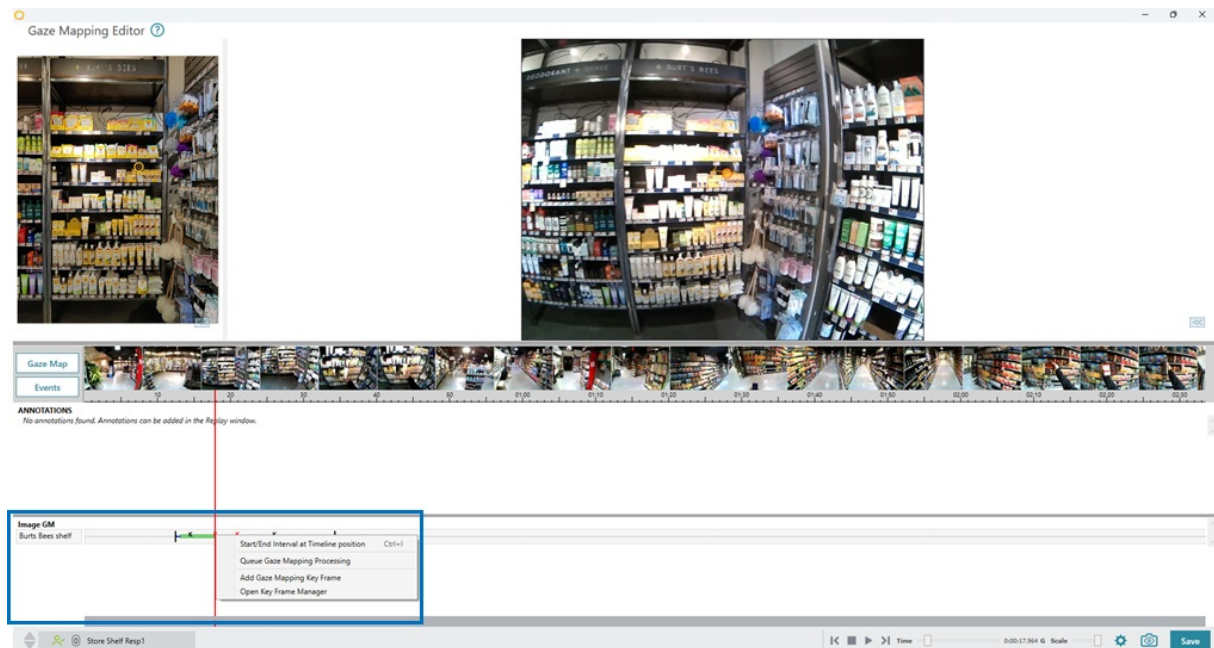
Reviewing Processed Gaze Maps

- You can review Gaze Maps by comparing the eye tracking data that appears in the Reference Image (top left) and Stimulus Recording (top right).
- Eye tracking data should appear in the Reference Image when it appears in the Stimulus Recording - gaze data between these images should align as expected (based on visual inspection of expected locations in stimulus recording).
- For each Gaze Map timeline:
 - Green indicates where gaze mapping was applied, i.e. where the reference images was observed.
 - Blue indicates where gaze mapping was processed but not applied - take note if the Gaze Map should apply so that Key Frame can be applied (e.g. when the reference image appears in stimulus recording).
 - Gray indicates gaze maps/intervals that require processing
 - To delete a Gaze Map, right-click on the name in the list on the left of the 'Image GM' panel - then choose to delete for the current respondent or all respondents.

If needed, Gaze Maps can be refined with Key Frames (described next).

Creating a Key Frames

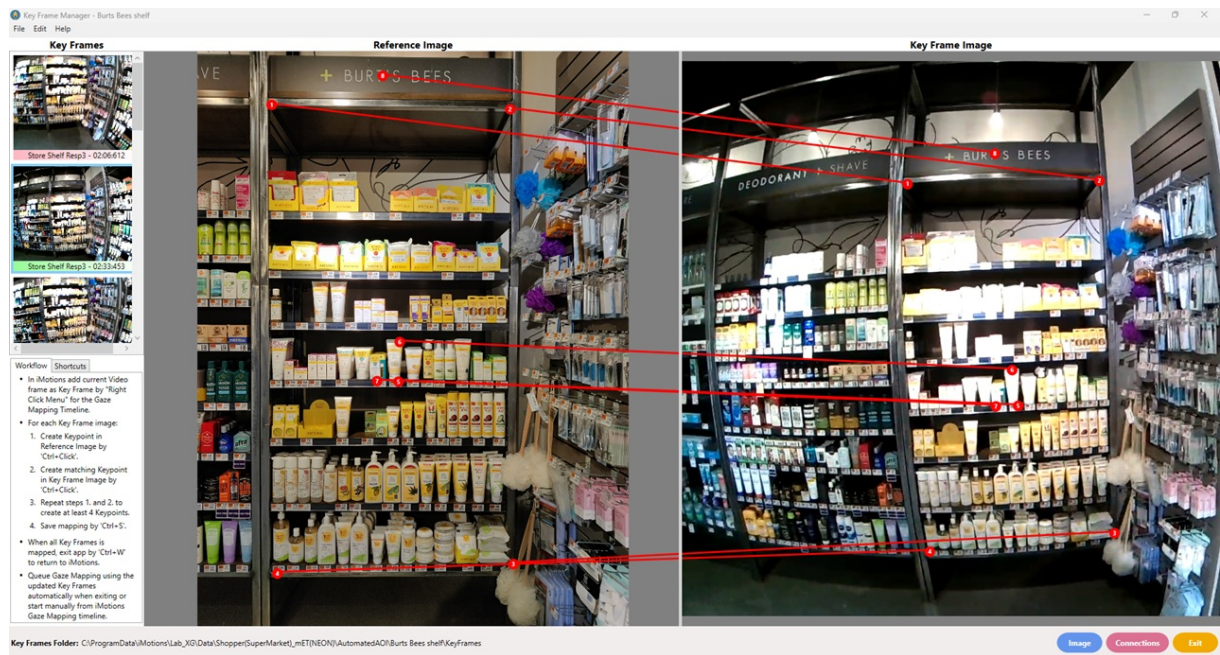
1. Navigate to time point for creating a Key Frame (e.g. when the reference image appears in the stimulus recording but was not recognized/processed by in the gaze map).
2. Right-click on the Gaze Map timeline and select 'Add Gaze Mapping Key Frame'.
3. New Key Frames appear as a red 'K', until Keypoints are defined (described next).
4. Key Frames that have Keypoints appear as a black 'K'.



Key Frame options in Image GM timeline

Defining Key Frames with Keypoints

1. Right-click on the Gaze Map timeline and select 'Open Key Frame Manager'.
2. Refer to the Workflow and Shortcuts provided in the Key Frame Manager (bottom-left) to define 'Keypoints'.
3. Keypoints consist of a point in the reference image and the associated point in the stimulus recording.
4. We recommend defining at least 4 Keypoints per Keyframe and creating additional Keypoints as needed.
5. Once all Keyframes have been added, re-run Gaze Map processing.



Key Frame Manager